UNIT- V

NETWORK SECURITY AND EMERGING TECHNOLOGIES

Network Security Technologies: Intrusion detection systems, firewalls, VPNs. Blockchain Technology in Networking: Basics of blockchain, applications in networking. Zero Trust Networking: Concepts and implementation strategies

Intrusion Detection System

A system called an intrusion detection system (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is software that checks a network or system for malicious activities or policy violations. Each illegal activity or violation is often recorded either centrally using an SIEM system or notified to an administration. IDS monitors a network or system for malicious activity and protects a computer network from unauthorized access from users, including perhaps insiders. The intrusion detector learning task is to build a predictive model (i.e. a classifier) capable of distinguishing between 'bad connections' (intrusion/attacks) and 'good (normal) connections'.

Working of Intrusion Detection System(IDS)

- An IDS (Intrusion Detection System) monitors the traffic on a computer network to detect any suspicious activity.
- It analyzes the data flowing through the network to look for patterns and signs of abnormal behavior.
- The IDS compares the network activity to a set of predefined rules and patterns to identify any activity that might indicate an attack or intrusion.
- If the IDS detects something that matches one of these rules or patterns, it sends an alert to the system administrator.
- The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.

Classification of Intrusion Detection System(IDS)

Intrusion Detection System are classified into 5 types:

- **Network Intrusion Detection System (NIDS):** Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet and matches the traffic that is passed on the subnets to the collection of known attacks. Once an attack is identified or abnormal behavior is observed, the alert can be sent to the administrator. An example of a NIDS is installing it on the subnet where firewalls are located in order to see if someone is trying to crack the firewall.

- **Host Intrusion Detection System (HIDS):** Host intrusion detection systems (HIDS) run on independent hosts or devices on the network. A HIDS monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected. It takes a snapshot of existing system files and compares it with the previous snapshot. If the analytical system files were edited or deleted, an alert is sent to the administrator to investigate. An example of HIDS usage can be seen on mission-critical machines, which are not expected to change their layout.
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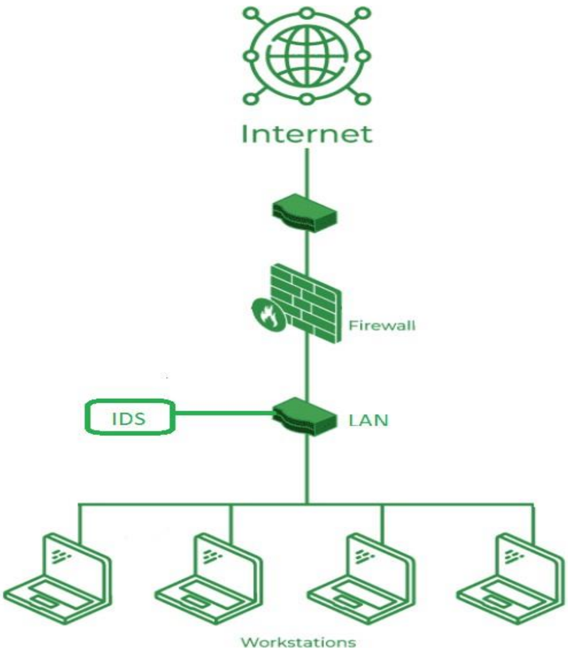


Intrusion Detection System (IDS)

- **Protocol-Based Intrusion Detection System (PIDS):** Protocol-based intrusion detection system (PIDS) comprises a system or agent that would consistently reside at the front end of a server, controlling and interpreting the protocol between a user/device and the server. It is trying to secure the web server by regularly monitoring the HTTPS protocol stream and accepting the related HTTP protocol. As HTTPS is unencrypted and before instantly entering its web presentation layer then this system would need to reside in this interface, between to use the HTTPS.

- **Application Protocol-Based Intrusion Detection System (APIDS):** An application Protocol-based Intrusion Detection System (APIDS) is a system or agent that generally resides within a group of servers. It identifies the intrusions by monitoring and interpreting the communication on application-specific protocols. For example, this would monitor the SQL protocol explicitly to the middleware as it transacts with the database in the web server.
- **Hybrid Intrusion Detection System:** Hybrid intrusion detection system is made by the combination of two or more approaches to the intrusion detection system. In the hybrid intrusion detection system, the host agent or system data is combined with network information to develop a complete view of the network system. The hybrid intrusion detection system is more effective in comparison to the other intrusion detection system. Prelude is an example of Hybrid IDS.

What is an Intrusion in Cybersecurity?

Understanding Intrusion Intrusion is when an attacker gets unauthorized access to a device, network, or system. Cyber criminals use advanced techniques to sneak into organizations without being detected. Common methods include:

- **Address Spoofing:** Hiding the source of an attack by using fake, misconfigured, or unsecured proxy servers, making it hard to identify the attacker.
- **Fragmentation**: Sending data in small pieces to slip past detection systems.
- **Pattern Evasion:** Changing attack methods to avoid detection by IDS systems that look for specific patterns.
- **Coordinated Attack:** Using multiple attackers or ports to scan a network, confusing the IDS and making it hard to see what is happening.

Intrusion Detection System Evasion Techniques

- **Fragmentation:** Dividing the packet into smaller packet called fragment and the process is known as fragmentation. This makes it impossible to identify an intrusion because there can't be a malware signature.
- **Packet Encoding:** Encoding packets using methods like Base64 or hexadecimal can hide malicious content from signature-based IDS.
- **Traffic Obfuscation:** By making message more complicated to interpret, obfuscation can be utilised to hide an attack and avoid detection.

- **Encryption:** Several security features, such as data integrity, confidentiality, and data privacy, are provided by encryption. Unfortunately, security features are used by malware developers to hide attacks and avoid detection.

Benefits of IDS

- **Detects Malicious Activity:** IDS can detect any suspicious activities and alert the system administrator before any significant damage is done.
- **Improves Network Performance:** IDS can identify any performance issues on the network, which can be addressed to improve network performance.
- **Compliance Requirements:** IDS can help in meeting compliance requirements by monitoring network activity and generating reports.
- **Provides Insights:** IDS generates valuable insights into network traffic, which can be used to identify any weaknesses and improve network security.

Detection Method of IDS

- **Signature-Based Method:** Signature-based IDS detects the attacks on the basis of the specific patterns such as the number of bytes or a number of 1s or the number of 0s in the network traffic. It also detects on the basis of the already known malicious instruction sequence that is used by the malware. The detected patterns in the IDS are known as signatures. Signature-based IDS can easily detect the attacks whose pattern (signature) already exists in the system but it is quite difficult to detect new malware attacks as their pattern (signature) is not known.
- **Anomaly-Based Method:** Anomaly-based IDS was introduced to detect unknown malware attacks as new malware is developed rapidly. In anomaly-based IDS there is the use of machine learning to create a trustful activity model and anything coming is compared with that model and it is declared suspicious if it is not found in the model. The machine learning-based method has a better-generalized property in comparison to signature-based IDS as these models can be trained according to the applications and hardware configurations.

Comparison of IDS with Firewalls

IDS and firewall both are related to network security but an IDS differs from a firewall as a firewall looks outwardly for intrusions in order to stop them from happening. Firewalls restrict access between networks to prevent intrusion and if an attack is from inside the network it doesn't signal. An IDS describes a suspected intrusion once it has happened and then signals an alarm.

Why Are Intrusion Detection Systems (IDS) Important?

An Intrusion Detection System (IDS) adds extra protection to your cybersecurity setup, making it very important. It works with your other security tools to catch threats that get past your main defenses. So, if your main system misses something, the IDS will alert you to the threat.

Advantages

- **Early Threat Detection**: IDS identifies potential threats early, allowing for quicker response to prevent damage.
- **Enhanced Security**: It adds an extra layer of security, complementing other cybersecurity measures to provide comprehensive protection.
- **Network Monitoring**: Continuously monitors network traffic for unusual activities, ensuring constant vigilance.
- **Detailed Alerts**: Provides detailed alerts and logs about suspicious activities, helping IT teams investigate and respond effectively.

Disadvantages

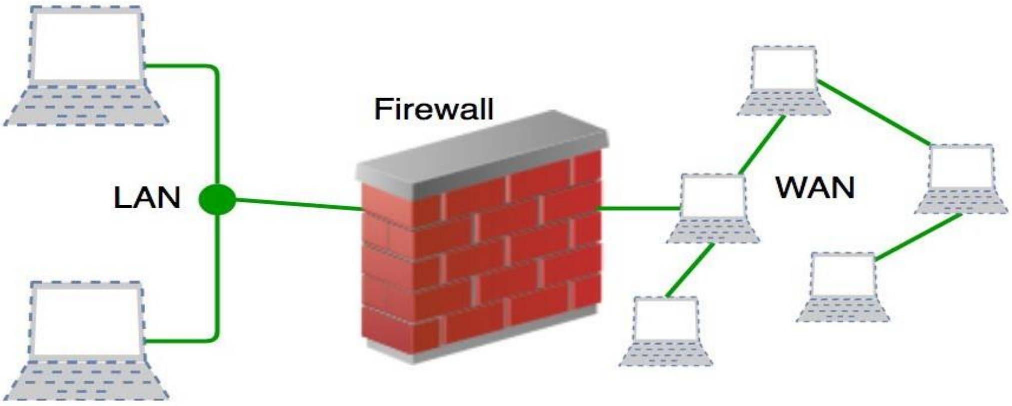
- **False Alarms**: IDS can generate false positives, alerting on harmless activities and causing unnecessary concern.
- **Resource Intensive**: It can use a lot of system resources, potentially slowing down network performance.
- **Requires Maintenance**: Regular updates and tuning are needed to keep the IDS effective, which can be time-consuming.
- **Doesn’t Prevent Attacks**: IDS detects and alerts but doesn’t stop attacks, so additional measures are still needed.
- **Complex to Manage**: Setting up and managing an IDS can be complex and may require specialized knowledge.

Firewall

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules accepts, rejects, or drops that specific traffic.

- **Accept:** allow the traffic
- **Reject:** block the traffic but reply with an "unreachable error"
- **Drop:** block the traffic with no reply

A firewall is a type of network security device that filters incoming and outgoing network traffic with security policies that have previously been set up inside an organization. A firewall is essentially the wall that separates a private internal network from the open Internet at its very basic level.



Working of Firewall

Firewall match the network traffic against the rule set defined in its table. Once the rule is matched, associate action is applied to the network traffic. For example, Rules are defined as any employee from Human Resources department cannot access the data from code server and at the same time another rule is defined like system administrator can access the data from both Human Resource and technical department. Rules can be defined on the firewall based on the necessity and security policies of the organization. From the perspective of a server, network traffic can be either outgoing or incoming.

Firewall maintains a distinct set of rules for both the cases. Mostly the outgoing traffic, originated from the server itself, allowed to pass. Still, setting a rule on outgoing traffic is always better in order to achieve more security and prevent unwanted communication. Incoming traffic is treated differently. Most traffic which reaches on the firewall is one of these three major Transport Layer protocols- TCP, UDP or ICMP. All these types have a source address and destination address. Also, TCP and UDP have port numbers. ICMP uses *type code* instead of port number which identifies purpose of that packet.

Default policy: It is very difficult to explicitly cover every possible rule on the firewall. For this reason, the firewall must always have a default policy. Default policy only consists of action (accept, reject or drop). Suppose no rule is defined about SSH connection to the server on the firewall. So, it will follow the default policy. If default policy on the firewall is set to *accept*, then any computer outside of your office can establish an SSH connection to the server. Therefore, setting default policy as *drop* (or reject) is always a good practice.

Types of Firewall

Firewalls can be categorized based on their generation.

1. Packet Filtering Firewall

Packet filtering firewall is used to control network access by monitoring outgoing and incoming packets and allowing them to pass or stop based on source and destination IP address, protocols, and ports. It analyses traffic at the transport protocol layer (but mainly uses first 3 layers). Packet firewalls treat each packet in isolation. They have no ability to tell whether a packet is part of an existing stream of traffic. Only It can allow or deny the packets based on unique packet headers. Packet filtering firewall maintains a filtering table that decides whether the packet will be forwarded or discarded. From the given filtering table, the packets will be filtered according to the following rules:

	Source IP	Dest. IP	Source Port	Dest. Port	Action
1	192.168.21.0	--	--	--	deny
2	--	--	--	23	deny
3	--	192.168.21.3	--	--	deny
4	--	192.168.21.0	--	>1023	Allow

Sample Packet Filter Firewall Rule

- Incoming packets from network 192.168.21.0 are blocked.
- Incoming packets destined for the internal TELNET server (port 23) are blocked.
- Incoming packets destined for host 192.168.21.3 are blocked.
- All well-known services to the network 192.168.21.0 are allowed.

2. Stateful Inspection Firewall

Stateful firewalls (performs Stateful Packet Inspection) are able to determine the connection state of packet, unlike Packet filtering firewall, which makes it more efficient. It keeps track of the state of networks connection travelling across it, such as TCP streams. So the filtering decisions would not only be based on defined rules, but also on packet's history in the state table.

3. Software Firewall

A software firewall is any firewall that is set up locally or on a cloud server. When it comes to controlling the inflow and outflow of data packets and limiting the number of networks that can be linked to a single device, they may be the most advantageous. But the problem with software firewall is they are time-consuming.

4. Hardware Firewall

They also go by the name "firewalls based on physical appliances." It guarantees that the malicious data is halted before it reaches the network endpoint that is in danger.

5. Application Layer Firewall

Application layer firewall can inspect and filter the packets on any OSI layer, up to the application layer. It has the ability to block specific content, also recognize when certain application and protocols (like HTTP, FTP) are being misused. In other words, Application layer firewalls are hosts that run proxy servers. A proxy firewall prevents the direct connection between either side of the firewall, each packet has to pass through the proxy.

6. Next Generation Firewalls (NGFW)

NGFW consists of Deep Packet Inspection, Application Inspection, SSL/SSH inspection and many functionalities to protect the network from these modern threats.

7. Proxy Service Firewall

This kind of firewall filters communications at the application layer, and protects the network. A proxy firewall acts as a gateway between two networks for a particular application.

8. Circuit Level Gateway Firewall

This works as the Sessions layer of the OSI Model's . This allows for the simultaneous setup of two Transmission Control Protocol (TCP) connections. It can effortlessly allow data packets to flow without using quite a lot of computing power. These firewalls are ineffective because they do not inspect data packets; if malware is found in a data packet, they will permit it to pass provided that TCP connections are established properly.

Functions of Firewall

- Every piece of data that enters or leaves a computer network must go via the firewall.
- If the data packets are safely routed via the firewall, all of the important data remains intact.
- A firewall logs each data packet that passes through it, enabling the user to keep track of all network activities.
- Since the data is stored safely inside the data packets, it cannot be altered.
- Every attempt for access to our operating system is examined by our firewall, which also blocks traffic from unidentified or undesired sources.

Importance of Firewalls

So, what does a firewall do and why is it important? Without protection, networks are vulnerable to any traffic trying to access your systems, whether it's harmful or not. That's why it's crucial to check all network traffic.

When you connect personal computers to other IT systems or the internet, it opens up many benefits like collaboration, resource sharing, and creativity. But it also exposes your network and devices to risks like hacking, identity theft, malware, and online fraud.

Once a malicious person finds your network, they can easily access and threaten it, especially with constant internet connections.

Using a firewall is essential for proactive protection against these risks. It helps users shield their networks from the worst dangers.

What Does Firewall Security Do?

A firewall serves as a security barrier for a network, narrowing the attack surface to a single point of contact. Instead of every device on a network being exposed to the internet, all traffic must first go through the firewall. This way, the firewall can filter and block non-permitted traffic, whether it's coming in or going out. Additionally, firewalls help create a record of attempted connections, improving security awareness.

What Can Firewalls Protect Against?

- **Infiltration by Malicious Actors:** Firewalls can block suspicious connections, preventing eavesdropping and advanced persistent threats (APTs).
- **Parental Controls:** Parents can use firewalls to block their children from accessing explicit web content.

- **Workplace Web Browsing Restrictions:** Employers can restrict employees from using the company network to access certain services and websites, like social media.
- **Nationally Controlled Intranet:** Governments can block access to certain web content and services that conflict with national policies or values.

By allowing network owners to set specific rules, firewalls offer customizable protection for various scenarios, enhancing overall network security.

Advantages of Using Firewall

- **Protection From Unauthorized Access:** Firewalls can be set up to restrict incoming traffic from particular IP addresses or networks, preventing hackers or other malicious actors from easily accessing a network or system. Protection from unwanted access.
- **Prevention of Malware and Other Threats:** Malware and other threat prevention: Firewalls can be set up to block traffic linked to known malware or other security concerns, assisting in the defense against these kinds of attacks.
- **Control of Network Access:** By limiting access to specified individuals or groups for particular servers or applications, firewalls can be used to restrict access to particular network resources or services.
- **Monitoring of Network Activity:** Firewalls can be set up to record and keep track of all network activity.
- **Regulation Compliance:** Many industries are bound by rules that demand the usage of firewalls or other security measures.
- **Network Segmentation:** By using firewalls to split up a bigger network into smaller subnets, the attack surface is reduced and the security level is raised.

Disadvantages of Using Firewall

- **Complexity:** Setting up and keeping up a firewall can be time-consuming and difficult, especially for bigger networks or companies with a wide variety of users and devices.
- **Limited Visibility:** Firewalls may not be able to identify or stop security risks that operate at other levels, such as the application or endpoint level, because they can only observe and manage traffic at the network level.
- **False Sense of Security:** Some businesses may place an excessive amount of reliance on their firewall and disregard other crucial security measures like endpoint security or intrusion detection systems.

- **Limited adaptability:** Because firewalls are frequently rule-based, they might not be able to respond to fresh security threats.
- **Performance Impact:** Network performance can be significantly impacted by firewalls, particularly if they are set up to analyze or manage a lot of traffic.
- **Limited Scalability:** Because firewalls are only able to secure one network, businesses that have several networks must deploy many firewalls, which can be expensive.
- **Limited VPN support:** Some firewalls might not allow complex VPN features like split tunneling, which could restrict the experience of a remote worker.
- **Cost:** Purchasing many devices or add-on features for a firewall system can be expensive, especially for businesses.

VPN

A virtual private network (VPN) is a technology that creates a safe and encrypted connection over a less secure network, such as the Internet. A Virtual Private Network is a way to extend a private network using a public network such as the Internet. The name only suggests that it is a "Virtual Private Network", i.e. user can be part of a local network sitting at a remote location. It makes use of tuneling protocols to establish a secure connection.

History of VPNs

ARPANET introduced the idea of connecting distant computers in the 1960s. The foundation for current internet connectivity was established by ensuring the development of protocols like TCP/IP in the 1980s. Particular VPN technologies first appeared in the 1990s in response to the growing concerns about online privacy and security.

Need for VPN

It could easily be said that VPNs are a necessity since privacy, security, and free internet access should be everybody's right. First, they establish secure access to the corporate networks for remote users; then, they secure the data during the transmission and, finally, they help users to avoid geo-blocking and censorship. VPNs are highly useful for protecting data on open Wi-Fi, for privacy, and preventing one's ISP from throttling one's internet connection.

Characteristics of VPN

- **Encryption:** VPNs employ several encryption standards to maintain the confidentiality of the transmitted data and, even if intercepted, can't be understood.
- **Anonymity:** Thus, VPN effectively hides the users IP address, thus offering anonymity and making tracking by websites or other third parties impossible.
- **Remote Access:** VPNs provide the means for secure remote connection to business' networks thus fostering employee productivity through remote working.
- **Geo-Spoofing:** The user can also change the IP address to another country using the VPN hence breaking the regional restrictions of some sites.
- **Data Integrity:** VPNs make sure that the data communicated in the network in the exact form and not manipulated in any way.

Types of VPN

There are several types of VPN and these are vary from specific requirement in computer network. Some of the VPN are as follows:

- Remote Access VPN
- Site to Site VPN
- Cloud VPN
- Mobile VPN
- SSL VPN

VPN Protocols

- **OpenVPN:** A cryptographic protocol that prioritises security is called OpenVPN. OpenVPN is compatible protocol that provides a variety of setup choices.
- **Point-To-Point Tunneling Protocol(PPTP):** PPTP is not utilized because there are many other secure choices with higher and more advanced encryption that protect data.
- **WireGuard:** Wireguard is a good choice that indicates capability in terms of performance.
- **Secure Socket Tunneling Protocol(SSTP):** SSTP is developed for Windows users by Microsoft. It is not widely used due to the lack of connectivity.
- **Layer 2 Tunneling Protocol(L2TP)** It connects a user to the VPN server but lacks encryption hence it is frequently used with IPSec to offer connection, encryption, and security simultaneously.

Why Should Use VPN?

- **For Unlimited Streaming:** Love streaming your favourite shows and sports games? A VPN is your ultimate companion for unlocking streaming services.
- **For elevating your Gaming Experience:** Unleash your gaming potential with the added layer of security and convenience provided by a VPN. Defend yourself against vengeful competitors aiming to disrupt your gameplay while improving your ping for smoother, lag-free sessions. Additionally, gain access to exclusive games that may be restricted in your region, opening up a world of endless gaming possibilities.
- **For Anonymous Torrenting**: When it comes to downloading copyrighted content through torrenting, it's essential to keep your IP address hidden. A VPN can mask your identity and avoid potential exposure, ensuring a safe and private torrenting experience.
- **For supercharging your Internet Speed:** Are you tired of your Internet speed slowing down when downloading large files? Your Internet Service Provider (ISP) might be intentionally throttling your bandwidth. Thankfully, a VPN can rescue you by keeping your online activities anonymous, effectively preventing ISP throttling. Say goodbye to sluggish connections and embrace blazing-fast speeds.
- **Securing Public Wi-Fi:** VPNs are essential for maintaining security when using public Wi-Fi networks, such as those in coffee shops, airports, or hotels. These networks are often vulnerable to cyberattacks, and using a VPN encrypts your internet connection, protecting your data from potential hackers and eavesdroppers when you connect to untrusted Wi-Fi hotspots.

Tunnelling Protocols for VPN

- **OpenVPN:** An open source protocol with very good security and the ability to set up the functionality to use. Secure Sockets Layer / Transport Layer Security is for the key exchange; it can go through firewalls and network address translators (NATs).
- **Point-To-Point Tunneling Protocol (PPTP)**: Another outdated VPN protocol is PPTP as it is one of the oldest VPN protocols that are quite easy to configure but provides the weaker security than most contemporary VPN protocols.
- **WireGuard:** A relatively new protocol that has been widely recommended because of its relative ease of use and high performance. It incorporates modern techniques of encryption and it is perhaps easier to implement and to audit.

- **Secure Socket Tunnelling Protocol (SSTP):** SSTP is a Microsoft developed protocol; it is compatible with the Windows operating systems and uses SSL/TLS for encryption which is rather secure.
- **Layer 2 Tunnelling Protocol (L2TP):** L2TP is frequently combined with IPsec for encryption; however, L2TP does not have encryption integrated into it but does build a secure tunnel for data.

Authentication Mechanisms in VPN

- **Pre-Shared Key (PSK):** Is a secret key that is used for authenticating the two parties, that is, the client and the VPN server. It is easy to integrate but is also considered insecure when not administered properly.
- **Digital Certificates:** Based on certificates given by a reliable certificate authority, it is effective in identifying the identity of users and devices with a sense of security.
- **Username and Password:** Usually known in user authentication in which users submit their credentials for them to access the VPNs. This method is sometimes supported by other security measures such as MFA (multi-factor authentication).
- **Two-Factor Authentication (2FA):** Provides another level of protection by including a second factor of identification in the manner of a number received via one's cellular telephone along with a user identification and password.

Security Concerns in VPN

- **Data Leakage:** VPNs also can some time not hide IP address and thus cause leakages of data collected. This can happen via DNS leaks or when the VPN connection is severed prematurely or when switches between servers.
- **Weak Encryption:** Even the security of a VPN can be affected by weak encryption standards as well as outdated encryption algorithms. In this case, it is essential to implement sound encryption/decryption methods.
- **Trust in VPN Providers:** VPN provider can only guarantee that they will secure the user's data and refrain from abusing it if the user themselves trusts their service provider. Some providers may keep records of the use of the resource by a user and this can infringe on the privacy of a consumer.
- **Man-in-the-Middle Attacks (MitM):** If VPN setting is not safe, the attacker gets the chance to intervene and modify information exchanged between client and server.

- **Performance Trade-offs:** VPN security often affects internet connection since the encryption and routing through VPN servers cause slower connection. For security and performance are always equally important for the choice of the measures.

Benefits of VPN

- When you use VPN it is possible to switch IP.
- The internet connection is safe and encrypted with VPN
- Sharing files is confidential and secure.
- Your privacy is protected when using the internet.
- There is no longer a bandwidth restriction.
- It facilitates cost savings for internet shopping.

Limitations of VPN

- VPN may decrease your internet speed.
- Premium VPNs are not cheap.
- VPN usage may be banned in some nations.]

Blockchain

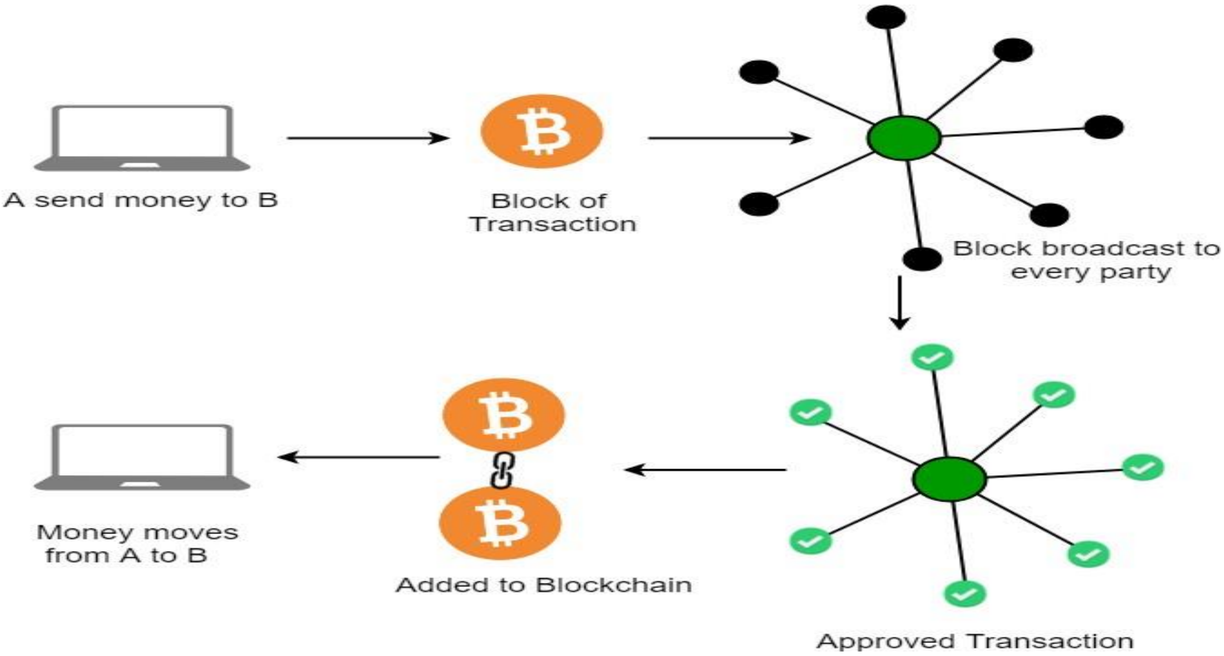
The blockchain is a distributed database of records of all transactions or digital events that have been executed and shared among participating parties. Each transaction is verified by the majority of participants of the system.

It contains every single record of each transaction. Bitcoin is the most popular cryptocurrency an example of the blockchain. Blockchain Technology first came to light when a person or group of individuals name 'Satoshi Nakamoto' published a white paper on "*BitCoin: A peer-to-peer electronic cash system*" in 2008.

Blockchain Technology Records Transaction in Digital Ledger which is distributed over the Network thus making it incorruptible. Anything of value like Land Assets, Cars, etc. can be recorded on Blockchain as a Transaction.

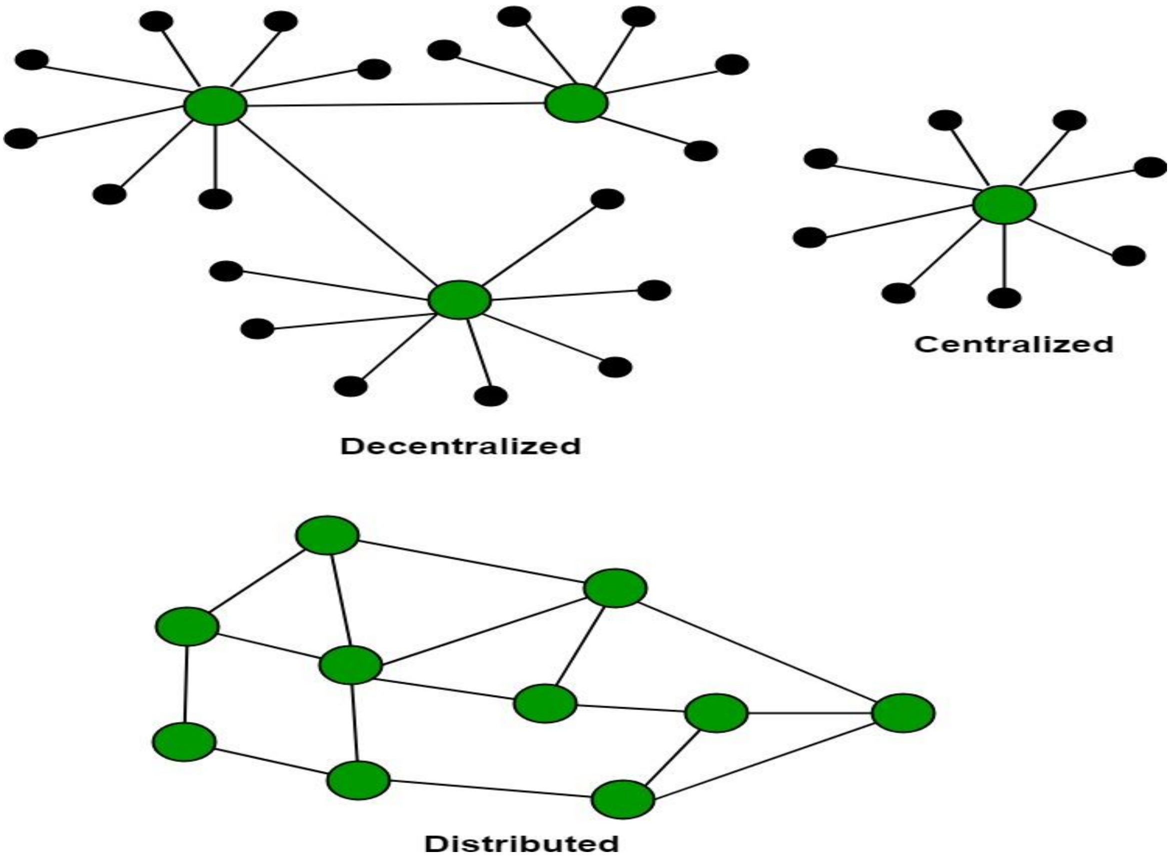
How does Blockchain Technology Work?

One of the famous use of Blockchain is Bitcoin. Bitcoin is a cryptocurrency and is used to exchange digital assets online. Bitcoin uses cryptographic proof instead of third-party trust for two parties to execute transactions over the Internet. Each transaction protects through a digital signature.



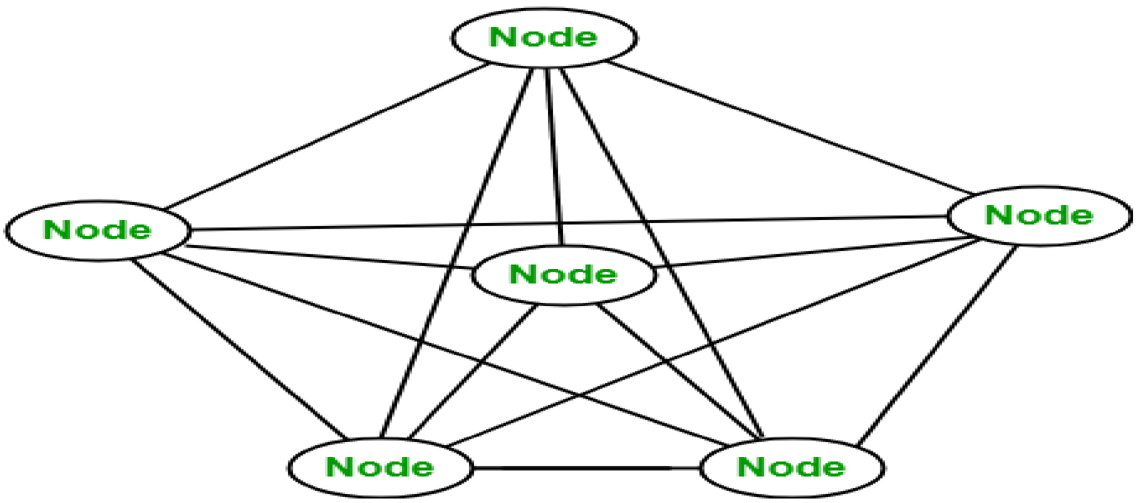
Blockchain Decentralization

There is no Central Server or System which keeps the data of the Blockchain. The data is distributed over Millions of Computers around the world which are connected to the Blockchain. This system allows the Notarization of Data as it is present on every Node and is publicly verifiable.



Blockchain nodes

A node is a computer connected to the Blockchain Network. Node gets connected with Blockchain using the client. The client helps in validating and propagating transactions onto the Blockchain. When a computer connects to the Blockchain, a copy of the Blockchain data gets downloaded into the system and the node comes in sync with the latest block of data on Blockchain. The Node connected to the Blockchain which helps in the execution of a Transaction in return for an incentive is called Miners.



Disadvantages of the current transaction system:

- Cash can only be used in low-amount transactions locally.
- The huge waiting time in the processing of transactions.
- The need for a third party for verification and execution of Transactions makes the process complex.
- If the Central Server like Banks is compromised, the whole system is affected including the participants.
- Organizations doing validation charge high process thus making the process expensive.

Building trust with Blockchain: Blockchain enhances trust across a business network. It's not that you can't trust those who you conduct business with it's that you don't need to when operating on a Blockchain network. Blockchain builds trust through the following five attributes:

- **Distributed:** The distributed ledger is shared and updated with every incoming transaction among the nodes connected to the Blockchain. All this is done in real time as there is no central server controlling the data.
- **Secure:** There is no unauthorized access to Blockchain made possible through Permissions and Cryptography.
- **Transparent:** Because every node or participant in Blockchain has a copy of the Blockchain data, they have access to all transaction data. They themselves can verify the identities without the need for mediators.
- **Consensus-based:** All relevant network participants must agree that a transaction is valid. This is achieved through the use of consensus algorithms.
- **Flexible:** Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Networks can evolve in pace with business processes.

What are the benefits of Blockchain?

- **Time-saving:** No central Authority verification is needed for settlements making the process faster and cheaper.
- **Cost-saving:** A Blockchain network reduces expenses in several ways. No need for third-party verification. Participants can share assets directly. Intermediaries are reduced. Transaction efforts are minimized as every participant has a copy of the shared ledger.
- **Tighter security:** No one can tamper with Blockchain Data as it is shared among millions of Participants. The system is safe against cybercrimes and Fraud.

- **Collaboration:** It permits every party to interact directly with one another while not requiring third-party negotiation.
- **Reliability:** Blockchain certifies and verifies the identities of every interested party. This removes double records, reducing rates and accelerating transactions.

Application of Blockchain

- Leading Investment Banking Companies like Credit Suisse, JP Morgan Chase, Goldman Sachs, and Citigroup have invested in Blockchain and are experimenting to improve the banking experience and secure it.
- Following the Banking Sector, the Accountants are following the same path. Accountancy involves extensive data, including financial statements spreadsheets containing lots of personal and institutional data. Therefore, accounting can be layered with blockchain to easily track confidential and sensitive data and reduce human error and fraud. Industry Experts from Deloitte, PwC, KPMG, and EY are proficiently working and using blockchain-based software.
- Booking a Flight requires sensitive data ranging from the passenger's name, credit card numbers, immigration details, identification, destinations, and sometimes even accommodation and travel information. So sensitive data can be secured using blockchain technology. Russian Airlines are working towards the same.
- Various industries, including hotel services, pay a significant amount ranging from 18-22% of their revenue to third-party agencies. Using blockchain, the involvement of the middleman is cut short and allows interaction directly with the consumer ensuring benefits to both parties. Winding Tree works extensively with Lufthansa, AirFrance, AirCanada, and Etihad Airways to cut short third-party operators charging high fees.
- Barclays uses Blockchain to streamline the Know Your Customer (KYC) and Fund Transfer processes while filling patents against these features.
- Visa uses Blockchain to deal with business-to-business payment services.
- Unilever uses Blockchain to track all their transactions in the supply chain and maintain the product's quality at every stage of the process.
- Walmart has been using Blockchain Technology for quite some time to keep track of their food items coming right from farmers to the customer. They let the customer check the product's history right from its origin.
- DHL and Accenture work together to track the origin of medicine until it reaches the consumer.

- Pfizer, an industry leader, has developed a blockchain system to keep track of and manage the inventory of medicines.
- The government of Dubai looking forward to making Dubai the first-ever city to rely on entirely and work using blockchain, even in their government office.
- Along with the above organizations, leading tech companies like Google, Microsoft, Amazon, IBM, Facebook, TCS, Oracle, Samsung, NVIDIA, Accenture, and PayPal, are working on Blockchain extensively.

Advantages of Blockchain Technology:

1. Decentralization: The decentralized nature of blockchain technology eliminates the need for intermediaries, reducing costs and increasing transparency.
2. Security: Transactions on a blockchain are secured through cryptography, making them virtually immune to hacking and fraud.
3. Transparency: Blockchain technology allows all parties in a transaction to have access to the same information, increasing transparency and reducing the potential for disputes.
4. Efficiency: Transactions on a blockchain can be processed quickly and efficiently, reducing the time and cost associated with traditional transactions.
5. Trust: The transparent and secure nature of blockchain technology can help to build trust between parties in a transaction.

Disadvantages of Blockchain Technology:

1. Scalability: The decentralized nature of blockchain technology can make it difficult to scale for large-scale applications.
2. Energy Consumption: The process of mining blockchain transactions requires significant amounts of computing power, which can lead to high energy consumption and environmental concerns.
3. Adoption: While the potential applications of blockchain technology are vast, adoption has been slow due to the technical complexity and lack of understanding of the technology.
4. Regulation: The regulatory framework around blockchain technology is still in its early stages, which can create uncertainty for businesses and investors.
5. Lack of Standards: The lack of standardized protocols and technologies can make it difficult for businesses to integrate blockchain technology into their existing systems.

6. Overall, the advantages of blockchain technology are significant and have the potential to revolutionize many industries. However, there are also several challenges and disadvantages that must be addressed before the technology can reach its full potential.

Zero Trust Networking (ZTN)

Zero Trust Networking (ZTN) is a modern security framework based on the principle of "never trust, always verify." Unlike traditional security models that rely on perimeter defenses, Zero Trust assumes that threats can exist both inside and outside the network. Here's a breakdown of the concepts and implementation strategies for Zero Trust Networking:

Core Concepts

1. **Never Trust, Always Verify**: Every request for access to resources is verified, regardless of where it originates (inside or outside the network).
2. **Least Privilege Access**: Users and devices are granted the minimum level of access necessary to perform their tasks. This limits potential damage if credentials are compromised.
3. **Micro-Segmentation**: The network is divided into smaller segments or zones. Access is controlled and monitored at each segment, reducing the attack surface.
4. **User and Device Authentication**: Both users and devices are authenticated before granting access. This often involves multi-factor authentication (MFA) and device health checks.
5. **Continuous Monitoring and Validation**: Continuous monitoring of user activity and device health is performed to detect and respond to anomalies in real-time.
6. **Explicit Access Controls**: Access to resources is explicitly defined and controlled based on a combination of user identity, device state, and other contextual factors.

Implementation Strategies

1. **Define the Protect Surface**: Identify and classify sensitive data, applications, and assets that need protection. This is often referred to as the "protect surface" and helps in focusing security efforts.
2. **Map the Transaction Flows**: Understand and document how data flows between users, applications, and devices. This helps in designing policies that control access based on these flows.
3. **Implement Micro-Segmentation**: Divide the network into segments and apply security policies to each segment. This limits lateral movement within the network and contains potential breaches.
4. **Enforce Least Privilege Access**: Implement role-based access control (RBAC) and attribute-based access control (ABAC) to ensure that users and devices have only the permissions necessary for their functions.
5. **Adopt Strong Authentication Mechanisms**: Use multi-factor authentication (MFA) and strong password policies. Ensure that devices are also authenticated and their compliance is verified.
6. **Continuous Monitoring and Analytics**: Deploy monitoring tools to continuously analyze network traffic, user behavior, and device health. Implement Security Information and Event Management (SIEM) systems to detect and respond to anomalies.
7. **Automate Response and Remediation**: Use automated tools and workflows to respond to detected threats and remediate issues quickly. This includes automated threat detection, alerting, and response actions.
8. **Zero Trust Network Access (ZTNA)**: Replace traditional VPNs with ZTNA solutions to ensure secure and granular access to applications without exposing the network to the public internet.
9. **Regularly Review and Update Policies**: Security policies should be regularly reviewed and updated to adapt to changing threats and organizational needs. Ensure that policies are enforced consistently.
10. **User and Device Training**: Educate users and administrators about Zero Trust principles and practices. Awareness and training are crucial for effective implementation and adherence.

Challenges and Considerations

- **Complexity**: Implementing Zero Trust can be complex, especially in large or legacy environments. A phased approach and careful planning are crucial.
- **Integration**: Integrating Zero Trust with existing systems and processes can be challenging. It requires coordination between different teams and technologies.
- **Cost**: The implementation of Zero Trust can involve significant upfront costs for new tools, technologies, and training.
- **Performance Impact**: Continuous verification and monitoring can impact network performance. Proper tuning and optimization are necessary to balance security and performance.
- **Change Management**: Adopting Zero Trust often involves changes to existing workflows and processes. Effective change management practices are essential to ensure a smooth transition.